EduGenie: Google Gemini Powered Learning Assistant

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1. Introduction

Running EduGenie locally means setting up the project on a computer so that the web interface and application can be developed and tested on the local machine before deployment. The local environment helps developers test the user interface, backend, Gemini integration, and learning features safely during development.

2. Requirements

- Computer or laptop with sufficient storage and memory.
- Modern web browser such as Google Chrome or Microsoft Edge.
- Code editor such as Visual Studio Code.
- Required runtime such as Python or Node.js, depending on the implementation.
- Stable internet connection for Gemini API access when the application uses the online Gemini service.
- A valid Gemini/Google AI API setup where required by the chosen implementation.

3. Project Folder Structure

```
EduGenie/
■■■ frontend/
■ ■■■ index.html
■ ■■■ style.css
■ ■■■ script.js
■■■ backend/
■ ■■■ app.py / server.js
■ ■■■ requirements.txt / package.json
■■■ README.md
```

4. Steps to Run EduGenie Locally

1. **Step 1 – Install Tools:** Install the required runtime, code editor, and project dependencies.
2. **Step 2 – Open Project:** Open the EduGenie project folder in the code editor.
3. **Step 3 – Install Dependencies:** Install the packages required by the frontend and backend.
4. **Step 4 – Configure Environment:** Store required configuration and API credentials securely using environment variables or the project's approved configuration method.
5. **Step 5 – Start Backend:** Start the local backend/API server.
6. **Step 6 – Start Frontend:** Start the frontend development server or serve the web files locally.
7. **Step 7 – Open Browser:** Open the local address shown by the development server.
8. **Step 8 – Test:** Enter a question and verify that the application displays the expected learning response.

5. Example Local Commands

For a typical Python backend:

```
python -m venv .venv
```

```
Activate the virtual environment
pip install -r requirements.txt
python app.py
```

For a typical Node.js project:

```
npm install
npm run dev
```

The exact commands depend on the project's technology and configuration.

6. Local Application Workflow

Browser → Local Frontend → Local Backend → Gemini Service → Local Backend → Frontend → Student

7. Testing Checklist

- Check that the home page loads correctly.
- Check that the question input accepts text.
- Check that the Ask/Submit button works.
- Check the loading indicator while a request is processed.
- Check that the AI response appears correctly.
- Test follow-up questions.
- Test invalid or empty input.
- Test the interface on mobile and desktop screen sizes.

8. Common Problems and Solutions

Problem	Possible Solution
Page does not open	Check that the local development server is running and use the correct local address.
Dependencies missing	Install the project's required packages using its dependency file.
AI response not received	Check the backend logs, configuration, network connection, and Gemini service setup.
API credential error	Verify the credential configuration and ensure secrets are not exposed in frontend code.
Port already in use	Stop the conflicting local process or configure the application to use another available port.

9. Security Note

API keys and other secrets should not be placed directly in client-side HTML, CSS, or JavaScript. For a web application, sensitive credentials should be handled by an appropriate backend or secure environment configuration.

10. Expected Result

After successful local setup, EduGenie can be opened in a browser through a local development address. Students can use the interface to enter learning questions, and the application can communicate with the configured Gemini service to generate learning assistance.